

- 2) Discuss the concept of product life cycle model. What are the various stages in product life cycle?

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- 3) Discuss the concept of economies of scale? How can economies of scale define current pattern of international trade in recent years?

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4.4 PRODUCT DIFFERENTIATION AND INTERNATIONAL TRADE

Currently, most of the international trade is based on differentiated product rather than homogeneous product in world market. For example, Audi is not identical to a Hyundai, Land Rover, Honda or a BMW. Therefore, most of the international trade occurred within differentiated products of the same industry or broad product group in same HS code² known as intra industry trade as opposed to inter-industry trade in completely different products. We have already discussed about Intra-industry trade in details in Unit 3. Let us discuss briefly here as a part of alternative explanations of trade theories. Intra- industry trade models are developed by various trade economists like Helpman, Krugman, Lancaster and others since 1979. Intra-industry trade or trade of similar products remains unexplained in the classical and Heckscher-Ohlin international trade models and theories. Comparative advantage theory earlier only explained the pattern of inter-industry trade. Intra- industry trade between countries occurred due to economies of scale in production. Competition within market forces each firm, plant or industry to focus on only few or one variety and styles of the same product rather than several diversified varieties and styles. This strategy is adopted by many firms to keep unit cost low as specialization decreases per unit cost of production. Thus, country can import the other varieties and styles from another country. Intra- industry trade lead to overall welfare of consumer as consumer received wider range of products at lower prices due to economies of scale in production. This can be explained with the example of European Union (EU). Before formation of EU, plant size of United States (US) firms and European firms was similar, but the unit cost were much higher in Europe due to many more varieties and styles of a product compared to US firm. When EU

²The Harmonized Commodity Description and Coding System, also known as the Harmonized System (HS) of tariff nomenclature is an internationally standardized system of names and numbers to classify traded products. It came into effect in 1988 and has since been developed and maintained by the World Customs Organization (WCO) (formerly the Customs Co-operation Council), an independent intergovernmental organization based in Brussels, Belgium, with over 200 member countries

formed and tariffs were reduced each plant specialized in the production of only few goods which sharply declined per unit cost in European Union.

The main reason for inter-industry trade is when we aggregate trade data at 2 digit level or HS-2 level; many products are aggregated into one category. For example, India imports and exports similar products like petroleum products, automobiles and so on. Petroleum products like crude oil, petrol, and diesel are aggregated into one category of product code under HS-2 code but actually India imports crude oil from Middle East countries which is processed in coastal areas of India and exported back to another country. Similarly, India imports components of automobiles from Japan as Japan has a comparative advantage in manufacturing of auto-components. These auto-components are assembled in India and exported back to another country. Most of the assembly lines and crude oil refineries are in India as labour is cheaper in India or India is labour-intensive country.

However, since all these types are generally aggregated into one export or import category, it could appear that the countries are exporting and importing “identical” products when they are actually exporting one type of automobile or petroleum products and importing another type. When there are no differences in resource endowments or technologies between countries’ economies of scale, discriminated products can explain intra-industry trade through is known as monopolistic competition model. Monopolistic competition model emphasise on consumer demand for a multiplicity of characteristics embodied in the goods sold in a same HS code category. In this model, trade is possible even when countries are very comparable in their productive capabilities.

The level of intra-industry trade can be measured by the intra- industry trade index by following formula:

$$T = 1 - \frac{|X - M|}{X + M}$$

Where T represent intra-industry trade index, X represent value of exports and M represents value of imports of a particular industry or commodity group and the vertical bars in the numerator of equation denote the absolute value. Intra industry trade index (T) value ranges from 0 to 1. T = 0 represent that a country only exports or imports the good or there is no intra industry trade and T=1 represent that intra- industry trade is maximum or exports and imports of a goods are equal. T index is calculated by various empirical research studies and found that current most of the trade is guided by intra- industry trade. In 1967, Grubel and Lloyd calculated T index for 10 industrial countries and found that combined weighted average of T for all industries in all 10 countries was 0.48. This result indicates that in 1967 nearly half of the trade among 10 industrial countries involved the exchange of differentiated products of the same industry.

4.5 GRAVITY MODEL OF TRADE

The Gravity Model of Trade is a key and robust empirical model in the international trade used to understand the pattern of international trade. This model was firstly introduced and used by Jan Tinbergen in 1962 employing gravity equation which is derived from Newton's theory of gravitation. As globes are attracted to each other in terms of their sizes and proximity similarly countries are also attracted by their sizes and proximities. According to gravity model, bilateral trade between two countries is proportional to size, measured by GDP, and inversely proportional to the geographic distance between them. The main focus of this model is to forecasts on the bilateral trade flows and these forecasts are based on the distance within two countries as well as their respective economic dimensions.

It is important to note that, the gravity model was not based on any theory but most of the trade economists empirically derived the impact of size and distance of the country on bilateral trade flows. As most of the trade economists used gravity model to explain pattern of trade therefore this model has been seen as a significant development on previous trade theories of Ricardo and Heckscher-Ohlin. Before introduction of gravity model none of the theoretical trade model used the size of an economy significantly in their explanation.

This model is important in terms of forecasting the impact of changes in trade policy on trade costs. This model is also flexible in terms of distance where it can cover different variables like tariff imposed at a border, cultural, institutional and political differences between trading nations. The Gravity model has recently used for understand the impact of BREXIT on trade flows between UK, EU and other countries.

Moreover, the gravity model is no longer concerned only with trade in goods, but has recently been applied with success to trade in services.

Intuitive gravity model:

Firstly gravity model was introduced as an intuitive way of understanding trade flows between countries. In its basic form, the gravity model can be written as follows:

$$\log X_{ij} = c + b_1 \log GDP_i + b_2 \log GDP_j + b_3 \log \tau_{ij} + e_{ij} \dots \dots \dots (1a)$$

$$\log \tau_{ij} = \log (\text{distance}_{ij}) \dots \dots \dots (1b)$$

In this model X_{ij} represent exports from country i to country j, GDP represent each country's gross domestic product, τ_{ij} indicates trade costs between the two countries, distance represent the geographical distance between two counties, as an observable proxy for trade costs and e_{ij} is a random error term. The c term is a regression constant, and the b terms are coefficients to be estimated.

The term “gravity” originates from the fact that the nonlinear form of equation 1a look like Newton’s law of gravity: bilateral trade between two countries is proportional to size, measured by GDP, and inversely proportional to the geographic distance between them. Putting into another words, gravity model explain that we expect similar countries trade more, but we assume countries wither larger distance trade less, possibly because transport costs between them are higher.

However, the intuitive gravity model has some limitation. For example, If India enters into a Free Trade Agreement with EU that lowers tariffs on their respective goods. Basic trade theory indicates that it will be also affect the trade with USA even though it is not party to the agreement. One of the important concepts of trade creation and trade diversion are examples of such effects. But intuitive gravity model does not account the impact of preferential trade agreement or free trade agreement on third country. This is clear from equation 1a which indicates that, $\frac{d \log x_{ij}}{d \log t_{ik}} = 0$

Therefore, decreasing trade costs on one bilateral route does not affect trade on other routes in the basic model, which contradicts standard trade theory.

A second problem with intuitive gravity model rises if we consider equal reductions in trade costs across all routes, including internal trade. For example there is decline in the price of crude oil, which drops transport costs everywhere, including inside countries. In the intuitive gravity model, this change would result in proportionate increases in trade across all bilateral routes, including internal trade. However, such an outcome is not considered in this model despite the change in trade costs as relative prices have not changed at all. In the absence of a change in relative prices, economic theory assumes that consumption patterns to remain constant for a given amount of total production measured by Gross Domestic Product.

As discussed in the previous sections there are problem with intuitive model of gravity hence this model was alerted by some trade economist. The first attempt in this regard was made by Anderson in 1979. However, the applied literature considered theoretical gravity model after the famous “gravity with gravitas” model of Anderson and Van Wincoop in 2003. This model is essentially a demand function which assumes that utility of consumers increases from consuming more variety of goods. From the production side, this model assumes that firm produces a single, unique product variety under increasing return to scale.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Critically discuss the concept of Intra-Industry Trade. How international trade is based on product differentiation?

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- 2) What is the Gravity model of trade? Explain with the help of formula and examples.

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4.6 KRUGMAN ALTERNATIVE THEORY OF TRADE

Krugman explained his theory of international trade by considering internal returns to scale and adopting a recent model of innovation under monopolistic competition. Internal economies of scale suggest that per unit average output cost will decrease as the total output increases. Most of the products have high fixed costs, and as production increases, it spreads the fixed cost of product. According to Krugman, if there are internal economies of scale, markets has fewer firms and each firm will produce more and discriminate their product from their competitors. Total number of firm is determined by average cost and price. When the price is higher than average, new firms will enter the market. Krugman also explained that larger economies have higher demand, hence these economies has higher internal return to scale.

Internal returns to scale is also applicable to International trade when there is more trade between countries. Suppose if trade between India and EU suddenly emerged due to free trade agreement then production in both markets will grow. There will be fewer total firms, and each persisting firm would produce more. Consumers in both countries would be able to buy variety of the products with low price. Per unit cost would also fall, because of the exploitation of further internal economies of scale. Therefore, even if there is no absolute or comparative advantage in trade, both countries will undertake trade to exploit the benefits of internal economies of scale. Countries will, all else same export the goods where market size or aggregate demand is highest or start their production process in a country where market size is higher by undertaking foreign direct investment. The main reason for this is that firms will be able to exploit higher internal economies of scale than other countries in that market. Therefore,

according to Krugman countries will tend to export the product in which they have specialization or they are able to produce more under the concept of internal economies of scale.

When tariff and non-tariff barriers to trade are more, factors of production will tend to move to countries where there are economies of scale in production. For example, suppose labour is only one factor of production, trade barriers would induce labour to move to the country with largest market as real wages will be more and there will be extra product diversity in that market. Krugman also explained his idea considering trade cost in the form of transaction costs. Under transaction costs specialization will be more limited, as trade costs decline the profitability of exporting. Severities of Internal economies of scale also decide the degree of specialization. Country will be specializing more when internal economies of scale will be more for products. Therefore, higher trade cost and limits of internal economies of scale lead to intra-industry trade.

According to Krugman trade between countries is also dependent on difference in factor endowments. Nation will export the good which are relatively abundant. But if two countries are equally abundant, it will lead to intra-industry trade between countries. If the countries are equally abundant, the effects for changes in the distribution of income as a result of trade then trade will be Pareto optimal. The gains from the trade and income distribution depend on the severity of difference between factor endowments. In case of more unique country's factor endowment, the more relatively scarce factor will lose from trade and the relatively abundant factor will gain. The international market scarce factor will lose from the trade because of the lower price of that product in that country as it faces competition from foreign producers who have lower costs because of relative abundance. Whether trade is Pareto optimal is determined by whether the benefit increase from product difference is great enough to make up for the comparative loss of income for the scarce factor.

Comparative advantage theory fail to explain trade based on internal economies of scale therefore Krugman explained the idea of internal economies of scale and monopolistic competition. If there are no product differences in relative costs, technology and tastes, gains from trade can be generated from lower prices and greater product diversity. This is an alternative approach to the division of labour and trade theory.

4.7 COST OF LOGISTICS, ENVIRONMENTAL STANDARDS, AND INTERNATIONAL TRADE

Classical trade theories assumed that there is no transportation cost or transportation cost is zero, but in actual case significant cost is needed for transporting good from one place to another place in international trade. Cost of transportation or logistics cost includes freight charges, warehousing costs, costs of loading and unloading, insurance premiums, and interest charges while goods

are in transit. Trade between two countries will occur only if the pre-trade price difference in the two countries exceeds the cost of transporting the good from one country to another country. Most of the commodities are not traded internationally due to high logistics costs. For example, Cement is not transacted internationally due to high transportation cost because of its high weight to value ratio. Cement is only traded to border areas due to high transportation cost.

Generally price of non-traded goods and services are determined by domestic demand and supply, while price of traded goods and services are determined by demand and supply in the world market. Currently, logistics cost between countries are significantly declined which resulted in increase in international trade. For example, Fish, Meat, fruits and vegetable found in many European countries in winters are shipped from country like India. In the past, due to high transportation cost and unavailability of refrigerator containers many perishable goods used to spoil in the transportation phase. Logistics costs are much higher in developing countries rather than developed countries and denote a significant obstacle to international trade for industries in developing countries.

Transportation cost can be analysed by two methods. One is by general equilibrium model and another by partial equilibrium analysis. Partial equilibrium analysis assumes constant exchange rate and constant level of income in two countries.

Logistics cost also influence the location of production under the international trade. Industries can be classified as market seeking, resource seeking or footloose industries. Resource seeking industries are establishing their production near to the source of raw material used by the industry. The main reason for this is as the transporting raw material used by the industry is substantially higher than for shipping the finished product to market. For example basic chemicals, aluminium and steel which processes heavy and bulky raw material into lighter finished products. On the other hand, market seeking industries established their plant near to the markets. Market seeking industries normally produce goods that become bulky to transport during the production process. For example, cold-drink industries like Coca-Cola, Pepsi established their manufacturing plant near to market as bottling weight is higher than their raw material like concentrated syrup. Finally footloose industries are producing the goods that face neither substantial weight gain or loss during the production process. These industries tend to be established near to availability of cheaper input cost. An example is provided by Japanese automobile industries which ship automobile machinery to India to be assembled due to cheap labour availability in India. Currently most of the countries are giving preferential tax treatment to attract these footloose industries.

International trade is also affected by environmental standards in different countries. Environmental standards refer to levels of air pollution, water pollution, thermal pollution and pollution resulting from garbage disposal that a

nation allows. International trade which creates negative impact on environment can lead to serious trade problems because the price of traded goods and services often does not reflect the negative externalities or socio environmental costs. Countries with lower environmental standards are able to attract more polluting industries. For example, United States labour opposed NAFTA due to fear that it would lead to job losses as industries might move to Mexico to take benefit of more relax or less environmental laws compare to U.S market. According to world bank study in 1992, exports of polluting industries expanded faster than clean industries in the last two decades in the world market and mostly shifted to underdeveloped and developing countries from developed nation (Low, 1992). As country make progress, people demand more clean industries rather than polluting industries. Most of the developed countries are now taking help of environmental standards to protect their economy from polluting industries which also increased non-tariff barriers amongst the countries.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Discuss the Krugman's alternative theory of international trade.

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2) Discuss the impact of environmental standards and logistics cost on location of the production and international trade.

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3) What is meant by resource- seeking industries? Market seeking industries? Footloose industries? What determine the classification of the industry? How does this affect the pattern of international trade?

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4.8 LET US SUM UP

In this unit, we discussed about alternative explanation of international trade theories. Classical theories of International Trade unexplained significant portion of real International trade flows between countries. The current unit fill this gap by providing alternative explanation of international trade which is based on empirical evidences of current trade flow with more relaxed and realistic assumptions to explain current patterns of international trade. Alternative explanation of trade theories are based on concept of increasing return to scale, economies of scale, product life cycle theory, Gravity Model, Product